

Crop Yield Optimizer Report

Paddy | BPT 5204 (Sona Masuri)

Plan Summary

Farmer ID:	farmer_001
Plan ID:	yopt_164a6def
Crop:	Paddy
Variety:	BPT 5204 (Sona Masuri)
Area:	2.5 acres
Stage:	planning
Confidence:	75% (model self-assessed)

Advisory report. All recommendations must be validated with local KVK officer before application.

1. Variety Selection & Land Preparation

BPT 5204 is the definitive choice for this 2.5-acre kharif paddy plot on black cotton vertisol with a single bore well: it commands a consistent Rs.2,200-2,800/quintal mandi premium in Telangana, has universal buyer acceptance across Hyderabad Monda Market and local mandis, and its 130-145 day duration fits the kharif window cleanly. The black cotton soil's native high-K content and moisture retention during early kharif directly benefit BPT 5204's yield ceiling, which sits at 4-5 t/acre under split-N + AWD management - well within reach on 2.5 acres at a Rs.60,000 investment cap. AWD (Alternate Wetting & Drying) is the prescribed water regime here: it saves 15-20% irrigation load on the single bore well without any yield penalty, which is critical given the lone water source and the soil's tendency toward waterlogging if continuously flooded.

Land Preparation

Soil test needed: Yes - get results before transplanting

Amendmen

2. Spacing, Water & Irrigation

Plant spacing: 0.25 m (row) x 0.25 m (plant)

Plant densi

Water Regime

Method: AWD

Single bore
yield penalt
transplanting
re-irrigate to
flood throug
drainage - n
AWD dry-do

Water savings vs flood: 28%

Monthly Irrigation Schedule		
Month	Depth (mm)	Frequency / Notes
June	80.0	Twice at transplanting establishment - initial flood to 5 cm on transplanting day, then AWD cycle begins; allow drain to 15 cm below surface before first re-irrigation (~5-7 days depending on bore well output and soil crack rate)
July	60.0	Every 7-9 days (AWD cycle) - monsoon rainfall supplements significantly; irrigate only when AWD tube shows water level at ?15 cm; expected 2-3 irrigation events if monsoon is active; skip entirely during heavy rain weeks
August	65.0	Every 7-10 days (AWD cycle) - active tillering through panicle initiation; increase vigilance on AWD tube readings; re-flood unconditionally at panicle initiation (?last week of August / first week of September for June-transplanted BPT 5204)
September	70.0	Every 5-7 days - maintain continuous shallow flood (3-5 cm) from panicle initiation through flowering and early grain fill; this is the highest water-demand window; bore well must run reliably; do NOT let field dry during this phase
October	50.0	Every 8-10 days - grain fill to dough stage; taper irrigation as grain matures; maintain 2-3 cm shallow water; begin final drainage at DAT 110-115 (10 days before targeted harvest)
November	0.0	No irrigation - field drained dry for combine access and grain hardening; harvest BPT 5204 at 20-22% grain moisture (target first two weeks of November for June-transplanted crop)

- Growth-Stage Overrides

- TRANSPLANTING (DAT 0): Flood field to 2-3 cm standing water on transplanting day to settle seedlings; drain to moist-only within 48 hours - do NOT keep flooded in first week (SRI protocol)

- ACTIVE every 10

3. Nutrition Program

Stage	DAP Range	Fertilizers Applied
Basal (Pre-Transplanting)	DAT 0 (at final puddling, 1-2 days before transplanting)	DAP 50 kg/acre - broadcast and incorporate into puddled soil; provides 23 kg P2O5 + 9 kg N per acre, MOP 35 kg/acre - broadcast and incorporate; provides 21 kg K2O; black cotton has moderate native K but BPT 5204 grain fill demands supplementation, ZnSO4 (Zinc Sulphate) 10 kg/acre - broadcast basal; near-universal Zn deficiency in Telangana black cotton paddy soils; correct now to avoid mid-season rescue foliar, Single Super Phosphate (SSP) 25 kg/acre - sulfur source (11.9% S); black cotton often S-deficient; also contributes 3 kg P2O5, FYM 2.5 tonnes/acre - already broadcast and incorporated in land prep; provides slow-release N + organic carbon + microbial activity; counts toward ~10-12 kg N equivalent over the season
Tillering Top-Dress	DAT 21-25 (active tillering, 3 weeks after transplanting)	Urea 55 kg/acre - provides 25.3 kg N (35% of 72 kg total urea-N target); broadcast on moist soil (NOT flooded); drain field 24 h before application, re-flood 24 h after to minimize volatilization, Leaf Color Chart (LCC) check MANDATORY before applying - apply only if LCC score <= 4 (pale green); if LCC = 5 (dark green), delay 5 days and re-check; saves 12-18 kg urea/acre on high-fertility black cotton, Foliar 2% urea spray (optional rescue): if plants show yellowing but field cannot be drained for broadcast, foliar 2% urea (20 g/L) at 200 L/acre as interim correction
Panicle Initiation Top-Dress	DAT 50-60 (panicle initiation - flag leaf just visible, stem feels slightly swollen at node)	Urea 40 kg/acre - provides 18.4 kg N (25% of total); broadcast on moist soil (drain 24 h before, re-flood 24 h after), ZnSO4 0.5% foliar spray - dissolve 500 g ZnSO4 in 100 L water; spray 200 L/acre on the crop canopy at early morning; corrects residual Zn deficiency that slows panicle development on black cotton, KH2PO4 0.5% foliar spray (optional, if any leaf reddening or K stress visible) - 500 g per 100 L water; spray 200 L/acre, Silicon (potassium silicate 2%) foliar or rice husk ash slurry poured into standing water - strengthens stems, reduces lodging risk in BPT 5204 (moderately susceptible to lodging at high N)
Booting / Pre-Flowering Top-Dress	DAT 70-80 (boot stage - panicle bulge visible inside flag leaf sheath)	Urea 25 kg/acre - provides 11.5 kg N (15% of total); SKIP entirely if LCC score > 4 at DAT 70 check; high N here increases blast susceptibility and BPH attraction in humid black cotton microclimates, Tricyclazole 75% WP 120 g/acre preventive spray - BPT 5204 is blast-susceptible; boot stage + humid August/September = highest blast risk window; mix with sticker and apply as fine mist on flag leaf, Propiconazole 25% EC 200 ml/acre - preventive false smut control (Ustilaginoidea virens increasingly common in high-N Telangana paddy); apply at 50% heading

DAT 85-100 (milk to
dough stage)

KH₂PO₄ 1% foliar - 1 kg per 100 L water; spray 200 L/acre; corrects late-season K deficiency (common in continuous-paddy black cotton plots); improves grain weight and chalkiness in BPT 5204, 0.5% ZnSO₄ foliar (second dose if Zn deficiency symptoms visible - pale striping on young leaves) - apply only if symptomatic; skip if crop looks healthy, Water management priority: maintain 2-3 cm shallow flood; do NOT let field crack in this stage - any water stress during grain fill reduces test weight and increases % chalky grains (critical for Sona Masuri mandi grade)

DAT 110-125 (grain
at 20-22% moisture,
80% panicles
golden)

No fertilizer inputs - all nutrient windows closed, Stop all irrigation DAT 110-115; open peripheral drainage channel outlet fully, Foliar ethephon (optional, 500 ppm) if uneven maturity across field - accelerates uniform ripening on staggered tillers; use only if >15% green panicles remain at DAT 118, Harvest at 20-22% grain moisture for combine; dry to 14% within 48 hours for mandi storage - BPT 5204 is susceptible to grain discoloration if left in field past 24% moisture in humid conditions

- Basal (Pre-Transplanting)- Incorporate all basal inputs in the final puddling pass. Do NOT apply full urea at basal - high N at transplanting promotes BPH and BLB on black cotton soils with high inherent fertility. The DAP + FYM combination adequately meets early N demand for 8-12 day SRI seedlings.

- Tillering uptake.

Nitrogen Split Protocol - 40.0 kg N / acre total		
Growth Stage	% of Total N	Day After Sowing (DAP)
basal	25.0%	DAP 0
tillering	35.0%	DAP 22
panicle_initiation	25.0%	DAP 55
booting	15.0%	DAP 75

Foliar Corrections

- ZnSO4 0.5% (500 g in 100 L water, spray 200 L/acre) at panicle initiation (DAT 50-60) - mandatory for black cotton Zn deficiency correction

- KH2PO4 chalkiness

4. IPM & Crop Protection

Monthly Pest Calendar

Month	Pest / Disease	Action Threshold	Organic First	Chemical (last resort)
January	No active crop (off-season)	N/A	Deep summer plowing prep; burn/bury previous crop residues to kill overwintering pupae and fungal inoculum.	None
February	No active crop (off-season)	N/A	Soil solarization (clear plastic mulch on tilled field) for 30 days if feasible - kills weed seeds, nematodes, and fungal spores. Procure pheromone trap lures and biocontrol agents.	None
March	No active crop (off-season)	N/A	Apply 2 tonnes FYM/acre (5 tonnes total) and incorporate into black cotton vertisol. Conduct soil health card test at KVK Suryapet - check pH (expect 7.5-8.5), available Zn, and P. Book certified BPT 5204 seed from KVK.	None
April	No active crop - land preparation	N/A	2-3 deep cross-plowings to break black cotton subsoil compaction and expose soil pathogens to UV. Prepare drainage channels along field perimeter - black cotton vertisol waterlogging risk is high under bore-well irrigation.	None
May	Pre-nursery soil pests (termites, root grubs)	Any termite activity in nursery bed area	Treat nursery bed soil with Trichoderma viride @ 2.5 kg/acre mixed with 50 kg FYM, broadcast and incorporate 1 week before nursery sowing. Apply neem cake @ 50 kg/acre to nursery bed.	If termite infestation confirmed in nursery bed: chlorpyrifos 20% EC @ 2.5 L/acre as soil drench (one-time pre-sowing application only).
June	Nursery - leaf folder (Cnaphalocrocis)			

medinalis) + thrips

Leaf folder: >5% leaf

damage in nursery; Thrips:
visible silvering on >10%
seedlings

NURSERY (sow 25-30 days

before transplant for
conventional; 8-12 days for
SRI). Spray NSKE (Neem Seed
Kernel Extract) 5% on nursery
beds every 7 days from Day 7
post-germination. Set up
pheromone traps for stem borer
(Scirpophaga incertulas) at 5
traps/acre in the main field now
- baseline monitoring.

Nursery thrips if

severe:
thiamethoxam
25% WG @ 0.2
g/L as a single
spray on nursery
seedlings 3-4
days before
uprooting for
transplant.

July

Yellow stem borer (*Scirpophaga*
incertulas) - dead heart stage; leaf
folder

Stem borer: >5
moths/pheromone trap/week
OR >5% dead hearts. Leaf
folder: >1 larva/tiller or
>10% leaf damage.

TRANSPLANTING MONTH.

Basal fertilizer: DAP 50 kg +
MOP 35 kg + 25% urea (35 kg)
incorporated before final
puddling. Release
Trichogramma japonicum egg
parasitoid @ 50,000 cards/acre
at 7 and 14 DAT. Spray Bt
(Bacillus thuringiensis var.
kurstaki) @ 1 g/L for leaf folder
at first sign.

Stem borer if threshold crossed:
chlorantraniliprol e 0.4% GR @ 4 kg/acre applied into standing water at 10-12 DAT (DO NOT use cartap GR - cartap hydrochloride 4% GR at 10 kg/acre is the alternative if chlorantraniliprol e unavailable).
Leaf folder if severe:
chlorantraniliprol e 18.5% SC @ 0.4 ml/L.

August

Brown Planthopper (BPH,
Nilaparvata lugens) - primary threat;
stem borer continued; sheath blight
(Rhizoctonia solani)

BPH: ≥ 10
nymphs+adults/hill (tap base
of 10 hills/acre twice
weekly). Stem borer: >5
moths/trap/week. Sheath
blight: visible lesions on
lower leaf sheaths
expanding.

ACTIVE TILLERING - MOST

CRITICAL MONTH. Apply first urea top-dress (35% of total N = ~50 kg urea/acre) at 21-25 DAT on MOIST (not flooded) soil - drain 24h before, re-flood 24h after. Use Leaf Color Chart (LCC): apply N only if LCC <= 4. Scout BPH twice weekly. Maintain alley-cutting (90 cm gap every 4 m) to improve air circulation and reduce BPH habitat.

BPH if threshold
crossed:
imidacloprid
17.8% SL @ 60
ml/acre OR
pymetrozine 50%
WG @ 120
g/acre. NEVER
use synthetic
pyrethroids
(lambda-cyhaloth
rin, cypermethrin)
- causes BPH
resurgence.
Sheath blight if
detected:
validamycin 3% L
@ 400 ml/acre,
repeat in 15
days. Stem
borer: cartap
hydrochloride 4%
GR @ 10 kg/acre
into standing
water.

BPH continued; blast (*Pyricularia oryzae*) - neck blast critical; false smut (*Ustilaginoidea virens*)

BPH: ≥ 10 /hill. Blast: any diamond-shaped gray-center lesions on leaves or at neck (zero tolerance for neck blast). False smut: greenish-yellow spore balls on any panicle.

PANICLE INITIATION (50-60 DAT) - Apply third N split (25% = ~35 kg urea/acre) + foliar ZnSO4 0.5% spray (corrects near-universal Zn deficiency on black cotton vertisol). Implement AWD (Alternate Wetting Drying): allow field to drain to 15 cm below soil surface between irrigations using perforated monitoring tube - saves 15-20% water on the single bore well. Scout for neck blast twice weekly from panicle initiation through heading.

Blast (preventive at boot stage, mandatory for BPT 5204 which is susceptible):
tricyclazole 75% WP @ 120 g/acre. If blast already visible:
tricyclazole immediately, followed by kasugamycin (rotate groups).
False smut preventive:
propiconazole 25% EC @ 200 ml/acre at booting. BPH rotation chemistry:
pymetrozine 50% WG @ 120 g/acre (if imidacloprid used in August - always rotate).

October

BPH (late instar); stem borer - white
ear phase; bacterial leaf blight
(Xanthomonas oryzae)

BPH: ≥ 10 /hill. Stem borer
white ears: $>1\%$ white
panicles. BLB: any linear
yellow-brown lesions from
leaf tip.

BOOTING (70-80 DAT) - Apply fourth and final N split (15% = ~20 kg urea/acre) ONLY if LCC <= 4 (skip if adequate). Foliar KH₂PO₄ 1% spray at grain fill to prevent late-season K deficiency (common in continuous paddy on black cotton). BEGIN field drainage 7-10 days before planned harvest for combine access.

BPH final
management:
dinotefuran 20%
SG @ 80 g/acre
(third rotation
chemistry). BLB
(no effective
curative): copper
oxychloride 50%
WP @ 600
g/acre as
bacteriostatic;
drain field and
reduce N. Stem
borer white ears:
nothing effective
at this stage -
document for
next season's
pheromone
threshold
calibration.

Storage pests (rice weevil *Sitophilus oryzae*, lesser grain borer)
post-harvest

Any visible weevil activity in stored grain within 2 weeks of storage

HARVEST TARGET MONTH
(BPT 5204, ~130-145 days
from sowing). Harvest at
20-22% grain moisture for
lowest breakage (<2%).
Sun-dry to 14% moisture before
storage. Line storage bags/bins
with neem leaves. Mix dry
neem leaf powder @ 1 kg/100
kg grain as storage protectant
(repels weevils).

If weevil
infestation
confirmed in
storage:
aluminium
phosphide
fumigation (3 g
tablet per tonne)
in sealed storage
- use only with
proper safety
equipment.
Prophylactic:
malathion 50%
EC dust @ 1
g/kg grain mixed
into storage (only
for
non-food-grade
stored paddy).

December

Post-harvest residue pests;
soil-borne pathogens from stubble

Any fungal discolouration on
stored paddy; stubble
decomposition stalled

Post-harvest crop residue management: incorporate stubble into soil with deep plowing OR burn stubble (burning recommended on black cotton vertisol when Sheath blight or BLB was detected during the season - kills inoculum). Apply Trichoderma viride @ 2.5 kg/acre mixed with FYM and broadcast on field to accelerate residue decomposition. Plan soil test for next season. Record all pest scouting data and yield for KVK reporting.

6. Harvest & Post-harvest

Maturity indicators:

80-85% of panicles showing golden-yellow color (grain ripening front moving from tip to base); Grain moisture at 20-22% - test with hand-squeeze: grain resists crushing but not rock-hard; 130-145 days after transplanting for BPT 5204 in Telangana kharif; Lower leaves dry and brown; upper flag leaf still partially green; Grain hard and translucent when bitten - milky stage is over

Harvest me

Post-harvest Steps

- Sun-dry immediately after threshing to reduce moisture from 22% to 14% (safe storage threshold) - spread on clean tarpaulin, 2-3 days in good sun

- Winnow

7. Economics & Cash Flow

Year-by-Year Yield Benchmarks				
Year	Low Yield	High Yield	Unit	Notes
Y1	20.0	26.0	quintals/acre	Year 1 kharif: Farmer learning SRI on black cotton vertisol. Lower bound = AWD + 4-split N but imperfect transplanting. Upper bound = full SRI protocol with young seedlings, 25x25 cm spacing.
Y2	23.0	28.0	quintals/acre	Year 2: Protocol familiarity improves; better seedling age discipline (8-12 days). Soil organic matter beginning to build from previous season residue incorporation.
Y3	25.0	30.0	quintals/acre	Year 3: SRI steady-state performance. Soil biology improving with green manure + residue management. Hitting SRI target range consistently.
Y4	25.0	32.0	quintals/acre	Year 4: Compounding soil health benefit. Zn and Si correction fully embedded. If switching to Rajamudi (red rice) intercycle, premium yield/revenue swaps in. BPT 5204 ceiling at ~30-32 q/acre under field conditions.
Y5	25.0	32.0	quintals/acre	Year 5: Plateau performance. Main variable is monsoon + BPH pressure. AWD water regime sustaining bore-well viability on black cotton site.
Y6	24.0	31.0	quintals/acre	Year 6: Slight risk of soil fatigue in continuous paddy; recommend introducing a rabi pulse (greengram/blackgram) or fallow rotation to refresh. Yield maintained if rotation observed.
Y7	24.0	31.0	quintals/acre	Year 7: Stable plateau. Si depletion possible in continuous paddy - apply rice husk ash 200 kg/acre to maintain stem strength and blast resistance.
Y8	23.0	30.0	quintals/acre	Year 8: Watch for bore-well water table decline (black cotton sites, single bore well). If water stress in August-September, yield loss 10-15%. Plan secondary water access.
Y9	22.0	30.0	quintals/acre	Year 9: Introduce soil health card-driven fertilizer revision. If pH has drifted alkaline

(>8.0 on black cotton), foliar Zn and Fe applications become more critical.

Y10

22.0

30.0

quintals/acre

Year 10: Recommend full crop rotation review at this stage - consider 1 year of cotton or pulses to break paddy disease cycles before Year 11 restart.

Cash Flow Projection				
Year	Investment (Rs.)	Revenue (Rs.)	Net (Rs.)	Notes
Y1	58,000	126,500	68,500	Year 1: Investm.
Y2	52,000	143,750	91,750	Year 2: Input c.
Y3	52,000	155,250	103,250	Year 3: SRI pla.
Y4	50,000	166,750	116,750	Year 4: Full ZB.
Y5	50,000	166,750	116,750	Year 5: Stable .
Y6	53,000	161,000	108,000	Year 6: Slight .
Y7	51,000	161,000	110,000	Year 7: Rice hu.
Y8	54,000	149,500	95,500	Year 8: Additio.
Y9	52,000	155,250	103,250	Year 9: Soil he.
Y10	51,000	155,250	104,250	Year 10: Recomm.

Production Cost Breakdown		
Cost Item	Annual Cost (Rs.)	Notes
Land preparation (2x plowing + puddling + leveling)	9,000	Tractor hire 2 passes Rs.2,500/acre × 2.5 = Rs.6,250 + puddling Rs.1,000/acre = Rs.2,500 + bund repair Rs.250. Total ~Rs.9,000.
Certified BPT 5204 seed (Year 1) / retained seed (Year 2+)	3,500	Year 1: 4 kg/acre × 2.5 acres × Rs.350/kg certified seed = Rs.3,500. SRI uses less seed than conventional (8-12 kg broadcast vs 4 kg for nursery). Year 2+ own retained seed ~Rs.500 cleaning cost only.
Fertilizer (DAP, Urea, MOP, ZnSO4) - 4-split schedule	14,000	DAP 50 kg/acre basal + Urea 4 splits total ~100 kg/acre + MOP 35 kg/acre + ZnSO4 5 kg/acre. For 2.5 acres: DAP 125 kg × Rs.27 = Rs.3,375 + Urea 250 kg × Rs.6.5 = Rs.1,625 + MOP 87.5 kg × Rs.17 = Rs.1,487 + ZnSO4 Rs.500 = ~Rs.7,000 chemical. + Jeevamruta inputs (jaggery, pulse flour, cow dung) Rs.1,500 for 2.5 acres. Total Rs.8,500 chemicals + Rs.1,500 bio = Rs.10,000. Add fungicide (tricyclazole, validamycin) + bactericide (copper OC)

Transplanting labor (SRI - 8-12 day seedlings, 25x25 cm spacing)

10,000

SRI transplanting is labor-intensive: 20-25 person-days/acre at Rs.400/day = Rs.8,000-10,000/acre × 2.5 = Rs.20,000-25,000. OFFSET: SRI saves nursery labor and skip conventional dense planting. Budget Rs.10,000 net transplanting labor for 2.5 acres.

Weeding (cono-weeder mechanical + manual topping)

5,000

Cono-weeder hire or own: 3 passes at 25 DAT, 35 DAT, 45 DAT. Labor 4 person-days/acre × Rs.400 × 2.5 acres = Rs.4,000. Herbicide (bispyribac-sodium if needed) Rs.1,000 for 2.5 acres.

Irrigation (bore-well power + operation)

5,000

AWD regime: ~12-14
irrigations vs 20+ in
continuous flood.
Electricity cost 5 HP
pump × 3 h/irrigation ×
13 irrigations × Rs.5/unit
= ~Rs.1,950. Plus diesel
backup + pump
maintenance Rs.3,000.
Total Rs.5,000/season
for 2.5 acres.

IPM (pheromone traps, biocontrol, targeted chemicals)

4,000

BPH
pheromone-equivalent
scouting kits +
Trichogramma japonicum
releases (3 × 50K/acre =
375K wasps ×
Rs.0.004/wasp =
Rs.1,500). Targeted
imidacloprid/pymetrozine
for BPH Rs.1,000.
Validamycin for sheath
blight Rs.500.
Miscellaneous IPM
supplies Rs.1,000.

Harvest (combine hire + post-harvest drying + storage bags)

5,500

Combine hire
Rs.1,200/acre × 2.5 =
Rs.3,000. Tarpaulin
drying labor Rs.500.
Hermetic storage bags (3
× 25 q bags) Rs.1,200.
Transport to procurement
center Rs.800. Total
Rs.5,500.

2,000

Soil test Rs.500 (every 3 years; annualized).
PMFBY insurance premium (1.5% of sum insured ~Rs.800).
Miscellaneous field supplies Rs.700.

Benchmark Comparison		
Metric	Value	Unit
Your target	27.5	quintals/acre
District average	20.0	quintals/acre
State best practice	32.0	quintals/acre

District average for BPT 5204 in Suryapet is 18-22 q/acre (conventional flood, single basal N, minimal IPM). This plan targets 25-30 q/acre (midpoint 27.5 q/acre) using full SRI protocol, 4-split N with LCC, AWD, and threshold-based IPM. The gap to district average is +7.5 q/acre (+37.5%), achievable without exotic technology - purely through management discipline. The gap to state best (32 q/acre, achievable under KVK demonstration plots with SRI + organic + zero BPH pressure) is -4.5 q/acre. This plan is realistically positioned at the 75th percentile of Telangana paddy performance, which is the correct ambition level for a farmer converting from conventional to SRI in Year 1 on a black cotton site with a single bore-well constraint.

8. Risk Register & Optimization Levers

Risk Register			
Risk	Probability	Impact	Mitigation
Brown Planthopper (BPH) outbreak - yield loss 30-70% if unmanaged	high	BPH at >10/hill can cause 'hopper burn' across the entire 2.5-acre field within 5-7 days. Black cotton vertisol with high native N + SRI dense canopy creates BPH-favorable conditions. Potential loss: Rs.40,000-80,000 in a bad year.	Twice-weekly scouting from tillering. Threshold-based spray (imidacloprid 60 ml/acre ONLY at >=10 BPH/hill). NEVER use pyrethroids. AWD water regime reduces BPH habitat. LCC-guided N application prevents over-fertilization which attracts BPH.
Bore-well failure or water table decline in kharif peak demand (August)	medium	Single bore-well dependency on black cotton vertisol is the biggest structural risk for this farm. If bore-well fails or water table drops during panicle initiation (most water-sensitive stage), yield loss 40-60%. Crop may partially fail. Cost of emergency bore-well deepening: Rs.80,000-1,50,000.	AWD regime reduces irrigation demand by 15-20% vs continuous flood - preserves bore-well life. Monitor bore-well water level weekly from July. Maintain NREGA application for farm pond (1 acre-foot storage = emergency buffer). Register under PM-KUSUM for solar pump (subsidized) to reduce pump failure risk.
Blast disease (Pyricularia oryzae) - neck blast at panicle stage	medium	BPT 5204 is susceptible to blast. Neck blast (attacking the panicle neck at booting) is yield-catastrophic - can destroy 50-80% of panicles in an affected plot. Particularly dangerous in years with high humidity, cloudy weather, and heavy N application.	Preventive tricyclazole 75% WP at 120 g/acre at boot stage (70 DAT) - non-negotiable in BPT 5204 cultivation. Avoid late evening irrigation (leaf wetness overnight triggers blast). N splits using LCC prevent excess foliar N that fuels blast. Rotate to hexaconazole for second spray if symptoms appear.
Monsoon failure or irregular rainfall - kharif moisture deficit in July-August	medium	Suryapet district has ~731 mm annual rainfall but 70% falls June-September. A 15-day dry spell in August (active tillering / PI stage) with single bore-well and black cotton's delayed water	

release can cause yield loss of 20-30%. No MSP compensation for partial crop failure without PMFBY coverage.

Enroll in PMFBY before sowing deadline (premium 1.5% of sum insured; government subsidizes balance). AWD regime as buffer. If dry spell >10 days forecast, prioritize irrigation at

panicle initiation above all other stages. Farm pond (NREGA-eligible) provides 1 additional irrigation cycle.

MSP procurement bottleneck - delayed or partial government procurement

medium

Telangana Civil Supplies procurement is reliable for registered farmers but can have delays of 15-45 days post-harvest. Open market price at Suryapet mandi can drop to Rs.1,600-1,900/q during peak harvest glut (October-November). On 60 quintals, this means Rs.24,000-42,000 revenue loss vs MSP.

Register on TS eProcurement portal before harvest. Invest in hermetic storage (Rs.1,200 per 3 bags) to hold grain 30-60 days without quality loss while awaiting government procurement. Join or form an FPO for collective MSP negotiation and priority procurement slot.

Sheath blight (*Rhizoctonia solani*) escalation in dense SRI canopy

medium

SRI's wider spacing reduces but does not eliminate sheath blight risk on black cotton with high humidity. Severe infection causes 20-30% yield loss. Black cotton's moisture-retentive nature extends humid canopy period.

Validamycin 3% L at 400 ml/acre at active tillering; repeat 15 days later if lesions progress. Maintain cono-weeder alley pruning for canopy airflow. Avoid over-application of N (which inflates canopy density and sheath blight pressure).

Input cost inflation (fertilizer, diesel, labor) eroding net margin

high

Urea, DAP, and MOP prices in India have been volatile (2022-2024 saw 30-50% increases). Labor costs in Telangana rising 8-12% annually. If costs rise 15% while MSP rises 5%, net margin compresses by Rs.8,000-15,000/acre.

Transition to Jeevamruta-based ZBNF by Year 3 - replaces 30-50% synthetic fertilizer. Mechanical weeding (cono-weeder) reduces labor dependency. Participate in KCC at 4% interest to reduce input financing cost. Targeted IPM vs calendar spraying cuts pesticide cost 30-40%.

Optimization Levers (ranked by priority)

#	Lever	+Yield	Investment	Difficulty	Payback
1	4-Split Nitrogen Schedule with Leaf Color Chart (LCC) - replace single basal application	+20%	Rs.2,000 one-time (LCC cards Rs.50 from KVK; extra fertilizer application labor Rs.1,950)	easy	Same season - Rs.27,000-33,000 additional net revenue for Rs.2,000 investment. 13-16× ROI.
2	Full SRI protocol - 8-12 day seedlings, 25×25 cm single-plant spacing, cono-weeder mechanical weeding	+30%	Rs.8,000-10,000 additional labor vs conventional transplanting (first season only; normalizes Year 2+)	moderate	Year 1 - 30% yield uplift = +7-8 quintals/acre × 2.5 acres × Rs.2,300 = Rs.40,000-46,000 additional revenue. 4-5× ROI in Year 1.
3	AWD (Alternate Wetting-Drying) water regime - reduces irrigation demand 15-20% with zero yield penalty	+5%	Rs.500 (perforated PVC monitoring pipe per acre × 2.5 = Rs.1,250; subtract Rs.750 diesel/electricity savings)	easy	Immediate - bore-well longevity protection is the primary payback (Rs.80,000-1,50,000 avoided bore-well deepening cost), plus 5% yield gain from better root oxygenation.
4	Pheromone-based BPH and stem borer scouting + threshold spray (replace calendar pyrethroid sprays)	+15%	Rs.4,000/season (scouting labor + targeted chemical applications replacing blanket sprays)	easy	Same season - prevents 30-70% yield loss in BPH outbreak years. Targeted spray saves Rs.3,000-4,000 vs blanket spray regimes. Net cost near zero; protects Rs.40,000-80,000 in crop value.

Foliar ZnSO4 0.5% at panicle initiation + silicon (rice
husk ash 100 kg/acre)

+8%

Rs.1,500/season
(ZnSO4 500 g/acre
× 2.5 = Rs.625 +
rice husk ash
Rs.875 for 2.5
acres)

Same season - Zn deficiency is near-universal in Telangana black cotton. Correcting it adds 8% yield on average. Si strengthens stems (less lodging) and reduces blast entry. Rs.1,500 investment for Rs.13,000-16,000 additional revenue.

6

PMEBY crop insurance + ZBNF Jeevamruta transition
(Year 2-3) for input cost reduction

+0%

PMFBY
Rs.800/season
(1.5% of sum insured).
Jeevamruta setup
Rs.1,500
(materials: cow dung, urine, jaggery, besan, soil). Saves
Rs.4,000-6,000/season in synthetic fertilizer from Year 3

PMFBY: pays
back
catastrophically
in 1 bad
monsoon year
(replaces Rs.1
L+ crop loss).
Jeevamruta:
Rs.4,000-6,000
annual fertilizer
saving from
Year 3 onward
= 3-4× annual
investment.

FPO aggregation for MSP procurement priority and collective input purchasing

+5%

Rs.2,000-5,000
FPO membership
fee (one-time). No
recurring
incremental cost.

2 seasons -
collective
procurement
gets 5-8%
discount on
inputs. MSP
procurement
priority reduces
storage holding
risk. If FPO has
milling unit:
additional
Rs.800-1,200/t
onne milling
margin on 5-6
tonnes =
Rs.4,000-7,200
additional
annual income.

9. Devil's Advocate Critique

Overall confidence:

55%

Biggest yield gap driver:

Single bore well capacity on low-permeability black cotton vertisol - if output falls below 1,500 L/h, the AWD re-flood at panicle initiation (the unconditional re-flood window that determines spikelet number and grain fill) cannot be executed reliably, directly capping yield regardless of how well every other intervention is managed.

Why the target IS realistic

- AWD on black cotton vertisol is genuinely advantageous - the soil's low permeability slows drainage between irrigation cycles, meaning the single bore well is less likely to be overwhelmed. This soil-water interaction directly supports the AWD protocol and makes 27.5 q/acre achievable on limited water.

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Why the target might NOT be realistic

- The single bore well is the critical single point of failure. At AWD's most conservative estimate, a 2.5-acre paddy plot still needs 8-10 ML/season. If bore well output is below 1,500 L/h (a real risk on black cotton vertisols where recharge is slow), extended dry-down cycles risk crossing from AWD into actual drought stress at panicle initiation - the most yield-sensitive stage. There is NO fallback water source mentioned in the plan, and one pump failure or power cut during the PI re-flood window can cost 20-30% of grain.

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- multi-see...
- the yield...

10. Week-by-Week Activity Calendar

Generated from N-split protocol, irrigation schedule and IPM calendar. Adjust dates to your actual sowing/transplant date.

Wk	DAP	Growth Stage	Irrigation	Fertilizer / Nutrition	IPM / Scouting
1	0-6	Nursery prep	Twice at transplanting establishment - initial flood to 5 cm on transplanting day, then AWD cycle begins; allow drain to 15 cm below surface before first re-irrigation (~5-7 days depending on bore well output and soil crack rate) 80.0mm	N split 25.0% (basal)	Routine scout
2	7-13	Nursery / Seed bed	Twice at transplanting establishment - initial flood to 5 cm on transplanting day, then AWD cycle begins; allow drain to 15 cm below surface before first re-irrigation (~5-7 days depending on bore well output and soil crack rate) 80.0mm	-	Scout: No active crop (off-season) N/A
3	14-20	Nursery / Seedling growth	Twice at transplanting establishment - initial flood to 5 cm on transplanting day, then AWD cycle begins; allow drain to 15 cm below surface before first re-irrigation (~5-7 days depending on bore well output and soil crack rate) 80.0mm	-	Routine scout
4	21-27	Transplanting & establishment	Twice at transplanting establishment - initial flood to 5 cm on		

transplanting day,
then AWD cycle
begins; allow drain to
15 cm below surface
before first
re-irrigation (~5-7
days depending on
bore well output and
soil crack rate) |
80.0mm

N split 35.0% (tillering)	Routine scout
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Transplanting &
establishment

Every 7-9 days (AWD cycle) - monsoon rainfall supplements significantly; irrigate only when AWD tube shows water level at ≥ 15 cm; expected 2-3 irrigation events if monsoon is active; skip entirely during heavy rain weeks | 60.0mm

-

6

Active tillering

Every 7-9 days (AWD cycle) - monsoon rainfall supplements significantly; irrigate only when AWD tube shows water level at ?15 cm; expected 2-3 irrigation events if monsoon is active; skip entirely during heavy rain weeks | 60.0mm

-

Scout: No active crop
(off-season) | N/A

Every 7-9 days (AWD cycle) - monsoon rainfall supplements significantly; irrigate only when AWD tube shows water level at ≥ 15 cm; expected 2-3 irrigation events if monsoon is active; skip entirely during heavy rain weeks | 60.0mm

-

8

Maximum tillering

Every 7-9 days (AWD cycle) - monsoon rainfall supplements significantly; irrigate only when AWD tube shows water level at ?15 cm; expected 2-3 irrigation events if monsoon is active; skip entirely during heavy rain weeks | 60.0mm

N split 25.0%
(panicle_initiation)

Every 7-10 days
(AWD cycle) - active
tillering through
panicle initiation;
increase vigilance on
AWD tube readings;
re-flood
unconditionally at
panicle initiation (?last
week of August / first
week of September
for June-transplanted
BPT 5204) | 65.0mm

-

10

Every 7-10 days
(AWD cycle) - active
tillering through
panicle initiation;
increase vigilance on
AWD tube readings;
re-flood
unconditionally at
panicle initiation (?last
week of August / first
week of September
for June-transplanted
BPT 5204) | 65.0mm

-

Scout: No active crop
(off-season) | N/A

Every 7-10 days
(AWD cycle) - active
tillering through
panicle initiation;
increase vigilance on
AWD tube readings;
re-flood
unconditionally at
panicle initiation (?last
week of August / first
week of September
for June-transplanted
BPT 5204) | 65.0mm

N split 15.0% (booting)

Every 7-10 days
(AWD cycle) - active
tillering through
panicle initiation;
increase vigilance on
AWD tube readings;
re-flood
unconditionally at
panicle initiation (?last
week of August / first
week of September
for June-transplanted
BPT 5204) | 65.0mm

-

Every 5-7 days -
maintain continuous
shallow flood (3-5 cm)
from panicle initiation
through flowering and
early grain fill; this is
the highest
water-demand
window; bore well
must run reliably; do
NOT let field dry
during this phase |
70.0mm

-

Every 5-7 days -
maintain continuous
shallow flood (3-5 cm)
from panicle initiation
through flowering and
early grain fill; this is
the highest
water-demand
window; bore well
must run reliably; do
NOT let field dry
during this phase |
70.0mm

-

Scout: No active crop - land
preparation | N/A

Every 5-7 days -
maintain continuous
shallow flood (3-5 cm)
from panicle initiation
through flowering and
early grain fill; this is
the highest
water-demand
window; bore well
must run reliably; do
NOT let field dry
during this phase |
70.0mm

-

Every 5-7 days -
maintain continuous
shallow flood (3-5 cm)
from panicle initiation
through flowering and
early grain fill; this is
the highest
water-demand
window; bore well
must run reliably; do
NOT let field dry
during this phase |
70.0mm

-

Every 8-10 days -
grain fill to dough
stage; taper irrigation
as grain matures;
maintain 2-3 cm
shallow water; begin
final drainage at DAT
110-115 (10 days
before targeted
harvest) | 50.0mm

-

Every 8-10 days -
grain fill to dough
stage; taper irrigation
as grain matures;
maintain 2-3 cm
shallow water; begin
final drainage at DAT
110-115 (10 days
before targeted
harvest) | 50.0mm

-

Scout: Pre-nursery soil pests
(termites, root grubs) | Any
termite activity in nursery bed
area

Key Milestones

- DAP 0 (Week 1): basal - apply 25.0% of total N (10.0 kg N/acre)
- DAP 22

Disclaimers & Data Sources

- This plan is advisory only - consult local KVK/ATMA before major investments.
- Prices are